# Who Moves First? Information Diffusion between Nasdaq-100 and S&P 500 Futures and ETFs, 2016–2026

*Validated lead-lag measures, macro announcements and the economics of a one-second lead*

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***ABSTRACT***

This paper measures how fast new information travels between the four most liquid US equity-index instruments: the E-mini Nasdaq-100 (NQ) and S&P 500 (ES) futures and the QQQ and SPY exchange-traded funds. It uses 9.4 million one-minute bars from May 2016 to October 2026, one-second data for 73–75 trading days in July–October 2026 and 453 scheduled macro announcements. Before estimating anything on real data, I show that standard Granger tests report a lead in 11 of 11 years when one market merely trades on a coarser price grid, and I validate an alternative *net lead* measure, built from generalized impulse responses, that recovers a planted 3-percentage-point lead to within 0.15 points. Three results follow. First, S&P 500 futures led Nasdaq-100 futures in 2016–2019, but the lead disappeared as NQ's share of the two markets' dollar volume rose from about 10% to 35% (t = 4.8). Second, futures lead their ETFs, clearly at one-second resolution (on 100% of days) but barely at one minute, and scheduled news reaches all four markets within the same minute. Third, the lead is economically tiny: at one-minute speed the follower's catch-up is at most 55% of the cheapest possible trading cost, and at one-second speed it roughly equals that cost and is gone within about two seconds. The lead survives exactly up to the cost of exploiting it, as Grossman and Stiglitz (1980) predict.

**Keywords:** price discovery, lead-lag, information diffusion, market microstructure, index futures, ETFs, event study.
**JEL:** G13, G14, C32, C58.

## 1. Introduction

When news arrives (an inflation print, a Federal Reserve decision, a large order from an informed trader) it does not reach every price at the same instant. Some markets absorb it first and others follow. The US equity-index complex is the natural place to study this: the Nasdaq-100 and the S&P 500 are each traded at the same time as a futures contract on the Chicago Mercantile Exchange (NQ and ES) and as an exchange-traded fund (QQQ and SPY). The four instruments carry largely the same information, so any delay between them is a direct measure of how information diffuses across markets.

The question matters for three groups. Traders and market makers need to know which market to watch when pricing the others. Hedgers and execution desks need to know whether a quote in one market is already stale when the other has moved. And economists care whether markets are informationally efficient: Grossman and Stiglitz (1980) argue that a predictable lead can survive only while it is smaller than the cost of trading on it. Hasbrouck (2003) found that index futures dominated price discovery over the corresponding ETFs in the early 2000s. Two decades of electronic trading, falling tick sizes relative to prices and a tenfold rise in NQ trading make it worth asking again.

The paper answers four questions. (i) Does one index future move first, and has this changed between 2016 and 2026? (ii) For each index, is information priced first in the future or in the ETF? (iii) How does diffusion change around scheduled macro announcements? (iv) Is any lead large enough to matter after trading costs, and for whom?

The contributions are three:

- **A methods check.** Using simulated markets and real NQ returns with a known, planted answer, I show that Granger causality tests are contaminated by differences in tick size, and that textbook price-discovery shares

cannot identify the leader at one-minute frequency. A *net lead* measure based on generalized impulse responses passes every test, so the conclusions rest on it.

- **A decade of evidence with a mechanism.** The ES-over-NQ lead of 2016–2019 faded as NQ became more liquid, and monthly estimates link the change to NQ's share of dollar volume.
- **An economic test.** Every statistical lead is translated into basis points that a trader could capture after the cheapest realistic cost, at one-minute and at one-second speed.

> **Main findings.** (1) ES led NQ by 1.86 percentage points in 2016–2019; since 2022 neither leads at one-minute resolution. (2) Futures lead their ETFs, and at one-second resolution NQ leads QQQ and ES leads SPY on every day of the sample. (3) Scheduled news is priced by all four markets in the same minute. (4) No lead can be traded profitably at human speed; at one-second speed the gain is about the size of the cost and lasts about two seconds.

Section 2 describes the data, Section 3 the methods and Section 4 the validation. Sections 5–9 present the results. Section 10 discusses limitations and Section 11 concludes.

## 2. Data

All market data come from London Strategic Edge: one-minute OHLCV bars for NQ, ES, QQQ and SPY from May 2016 to October 2026, one-second bars for the regular session (09:30–16:00 New York) from July to October 2026, and a US economic calendar with release dates, actual values and consensus forecasts. Using one vendor for every series gives consistent timestamps and cleaning. Table 1 summarises the sample.

**Table 1.** Sample.

| Instrument | Type | 1-minute bars | of which regular hours | Trading days | Seconds with a trade (1-s sample) |
|---|---|---|---|---|---|
| NQ | E-mini Nasdaq-100 future (CME) | 3,615,871 | 1,022,640 | 3,228 | 89.1% |
| ES | E-mini S&P 500 future (CME) | 3,639,152 | 1,026,097 | 3,231 | 83.0% |
| QQQ | Invesco QQQ Trust (Nasdaq-100 ETF) | 1,075,481 | 1,013,303 | 2,609 | 81.4% |
| SPY | SPDR S&P 500 ETF Trust | 1,074,269 | 1,013,285 | 2,610 | 84.0% |

One-minute sample: May 2016 – October 2026. Futures trade almost 23 hours a day, so their counts include the overnight session. One-second sample: 75 trading days for the futures and 73 for the ETFs, July–October 2026, regular hours only. Macro calendar: 122 CPI, 124 payroll, 125 ISM Manufacturing and 82 FOMC releases (453 in total).

**Cleaning.** Returns are one-minute log returns, $r_t = \ln(P_t/P_{t-1})$. Any return that spans a gap (overnight, weekend, the futures' daily maintenance break or a missing bar) is discarded, and all lags are taken on the clock ($t$ − 1 minute), never across a gap. Results are reported separately for regular hours and the futures' overnight session.

**Quality checks.** Futures and ETFs on the same index agree closely at the daily close: the daily return correlation is 0.995 for NQ–QQQ and 0.996 for ES–SPY, with a median absolute gap of about 1.3 basis points (bp). ETF closes match published historical closes, and the largest one-minute moves are genuine news (CPI releases at 08:30, the March 2020 emergency Fed cut, the April 2025 tariff pause), not data errors. Two vendor issues were found and handled. First, the economic calendar's timestamps are one hour late for about 90% of the back-filled history (Section 7). Second, futures data on 12 October 2026 start only at 14:43; the day is kept but shortened. An earlier version of the project used free Yahoo Finance data and was abandoned after checks found days on which an index and its ETF disagreed by more than 1%.

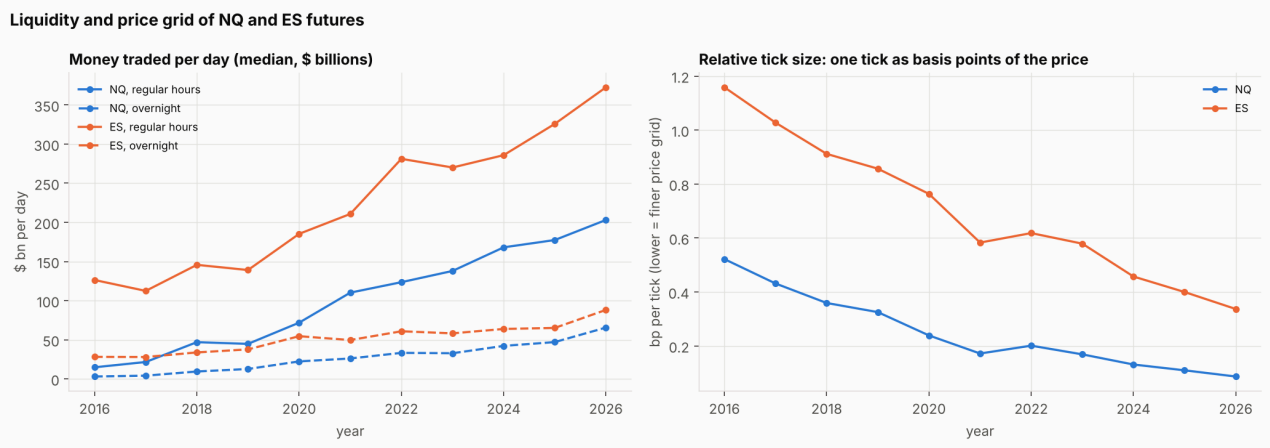



**Figure 1.** Market anatomy. Left: median dollar value traded per day. ES remains the larger market, but NQ's regular-hours turnover grew from about $15 bn a day in 2016 to about $203 bn in 2026, against ES's $126 bn to $373 bn. Right: one tick as basis points of the price. Both contracts move in 0.25-point steps, but because ES's price is lower, its tick is two to four times coarser relative to price (1.16 bp against 0.52 bp in 2016; 0.34 bp against 0.09 bp in 2026).

Two features of the market matter for what follows. NQ became far more liquid over the decade, and ES trades on a much coarser price grid relative to its price (Figure 1). A coarser grid means that small news moves show up in ES later, in a lump, when the price finally crosses a tick, even if nobody is slower to react. Section 4 shows that this alone is enough to fool a Granger test.

## 3. Methodology

### *3.1 Cross-correlation and Granger causality*

The starting point is the cross-correlation function $\rho(k) = \mathrm{corr}(x_t, y_{t+k})$. A positive correlation at $k > 0$ means that $x$ moves first. Significance uses the robust t-statistic of Dalla, Giraitis and Phillips (2022), which remains valid under volatility clustering, with a Bonferroni correction across lags. Granger (1969) causality is tested in the regression

$$y_t = c + \Sigma_{i=1\ldots p}\, a_i\, y_{t-i} + \Sigma_{i=1\ldots p}\, b_i\, x_{t-i} + \varepsilon_t, \quad H_0\colon b_1 = \ldots = b_p = 0 \tag{1}$$

with heteroskedasticity- and autocorrelation-consistent (HAC) standard errors and the lag length chosen by BIC.

### *3.2 Diffusion curve and net lead*

For each pair a bivariate VAR is estimated by OLS and its *generalized* impulse responses are computed (Pesaran and Shin, 1998). Unlike Cholesky-orthogonalised responses, they do not depend on which variable is ordered first, which matters when the same-minute correlation is about 0.9. The diffusion curve is the share of the long-run response of $y$ to a shock in $x$ that has arrived $h$ bars after the shock:

$$\mathrm{share}_{y\leftarrow x}(h) = \Sigma_{s=0\ldots h}\, \mathrm{GIRF}_{y\leftarrow x}(s) \;/\; \Sigma_{s=0\ldots H}\, \mathrm{GIRF}_{y\leftarrow x}(s) \tag{2}$$

One minus the share at $h = 0$ is the fraction of the reaction that arrives late. Microstructure noise such as bid-ask bounce makes the first bar overshoot in *both* directions, which biases each late share. The *net lead* takes the difference, so the common bias cancels:

$$\mathrm{net\ lead}_{\mathrm{NQ}} = [1 - \mathrm{share}_{\mathrm{ES}\leftarrow\mathrm{NQ}}(0)] - [1 - \mathrm{share}_{\mathrm{NQ}\leftarrow\mathrm{ES}}(0)] \tag{3}$$

It is measured in percentage points (pp). A positive value means that more of ES's reaction to NQ arrives late than the other way round, so NQ moves first.

### *3.3 Precision*

The net lead is estimated separately for every calendar month and session (about 125 months × 2). A period's estimate is the mean of its monthly estimates, with a 95% confidence interval of mean ± $1.96 \cdot \mathrm{sd}/\sqrt{n}$. This is the

logic of Fama–MacBeth standard errors: monthly estimates are close to independent even though minute-level returns are not.

### *3.4 Textbook price-discovery measures*

For pairs that track the same index, a vector error-correction model (VECM) is estimated each month on log prices. The error-correction term is the futures–ETF log gap, demeaned by day to remove the basis and dividends. From it come the Hasbrouck (1995) information share (both Cholesky orderings, giving upper and lower bounds), the Gonzalo–Granger (1995) component share and the Putniņš (2013) information leadership share. They are reported for comparison with the literature, and Section 4 tests whether they can be trusted.

### *3.5 Liquidity hypothesis*

To test whether the more liquid market leads, monthly net leads are regressed on NQ's share of NQ + ES dollar volume, with and without a linear time trend, using Newey–West standard errors with six lags.

### *3.6 One-second bars and the staleness trap*

One-second data are put on a common clock by previous-tick sampling: a market that did not trade in a second keeps its last price. That creates a *fake* lead, because the less active market's move appears a second later even if nobody reacted late (the non-synchronous trading problem behind the Epps (1979) effect). Lead-lag at one second is therefore measured only on seconds in which **both** markets traded, using the lag-1 asymmetry $\mathrm{corr}(a_t, b_{t+1}) - \mathrm{corr}(b_t, a_{t+1})$, estimated for each day and averaged across days with a 95% confidence interval. Section 8 checks the fix with a placebo.

### *3.7 Economic significance*

The test mimics what a lead-lag arbitrageur would do. When the leader $L$ makes one of its top-1% one-minute moves (ranked within each month), buy the follower $F$ in the same direction, hedge with β units of the leader and close after $h$ minutes. The hedge removes market-wide moves and leaves only the follower's catch-up:

$$\text{catch-up}(h) = \mathrm{sign}(r_{L,t}) \cdot [(p_{F,t+h} - p_{F,t}) - \beta(p_{L,t+h} - p_{L,t})], \quad \beta = \mathrm{cov}(r_F, r_L)/\mathrm{var}(r_L) \tag{4}$$

The catch-up is compared with the *cheapest* cost of the trade: one round trip in the follower at a one-tick spread, plus fees of about $2 per futures contract and $0.0035 per ETF share per side, with no market impact. These costs are deliberately optimistic, so a lead that does not beat them cannot be traded at that speed. Top-10% and top-0.1% moves are used as robustness checks. At one-second frequency the same trade is held for 1–10 seconds after the leader's top-0.1% seconds. There the gain is measured on the follower alone, because over longer holds the hedge would book the leader's reversal after its biggest seconds as a profit.

## 4. Validation: why the net lead, and not Granger

Before touching the real NQ–ES data, every measure was tested on data where the answer is known.

**Simulated markets.** In a toy market where 80% of each news shock reaches the follower in the same minute, the estimated same-minute share is 80.3%; with instant diffusion, no lead is found. On six simulated pairs of the same asset traded with different noise levels, the net lead had the right sign and size in every case.

**Real NQ noise.** Ten years of real one-minute NQ returns were used to build artificial followers whose true lead is known, so the test inherits the real data's fat tails, volatility clustering and intraday patterns. Table 2 shows the results.

**Table 2.** Validation on real NQ returns (11 years, regular / overnight hours).

| Scenario | Truth (pp) | Net lead found (pp) | Granger says "NQ leads" ($p < 0.01$) |
|---|---|---|---|
| Follower absorbs 3% of each move one minute late | +3.0 | **3.05 / 3.15** | 4 / 11 years |
| No lead | 0 | **0.04 / 0.06** | 0 / 11 and 3 / 11 years |
| No lead, but the follower trades on ES's coarser price grid | 0 | **−0.01 / −0.02** | **11 / 11 years (false alarm)** |

Net lead: mean across years of the estimate for the regular and the overnight session. In the tick-size scenario the follower is an ES-like price built from NQ (same information at the same instant) and rounded to a 0.25-point grid at ES's lower price level.

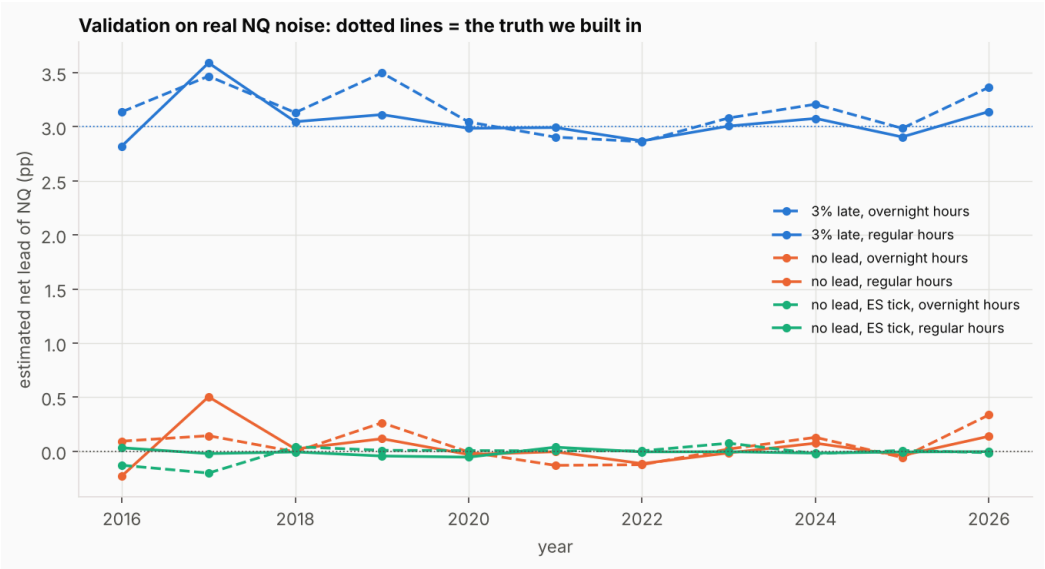



**Figure 2.** Net lead estimated on artificial pairs built from real NQ returns, by year. The dotted lines mark the truth that was built in: +3 pp (blue) and 0 (orange and green). The estimate tracks the truth in every year, including the scenario where the follower trades on a coarser price grid (green), the case in which Granger tests raise a false alarm every year.

The key result is the third row: **a coarser tick alone makes Granger tests report a lead that does not exist**, in every one of the eleven years. Because ES's tick is two to four times coarser than NQ's relative to price, Granger p-values on NQ versus ES cannot be trusted. The textbook VECM measures fail differently. On the simulated pairs, Hasbrouck's information share could only be bounded between about 0 and 1, and the component and leadership shares pointed the wrong way in several cases. The rest of the paper therefore draws its conclusions from the net lead.

## 5. Results I: NQ versus ES

**Table 3.** Net lead of NQ over ES (pp), mean of monthly estimates with 95% confidence interval. Negative values mean that ES moves first.

| Period | Months | Regular hours | Overnight |
|---|---|---|---|
| 2016–2019 | 43 | **−1.86** [−2.58, −1.14] | −1.65 [−2.54, −0.76] |
| 2020–2021 | 24 | +0.37 [−0.25, 0.99] | −0.46 [−1.19, 0.26] |
| 2022–2026 | 58 | **−0.10** [−0.38, 0.17] | −0.32 [−0.90, 0.26] |
| All years | 125 | −0.61 [−0.96, −0.27] | −0.81 [−1.25, −0.37] |

**The S&P futures used to move first; today neither does.** In 2016–2019 about 1.9 pp more of NQ's reaction to ES arrived a minute late than the other way round ($t = -5.1$). The lead then faded: in 2020–2021 the interval includes zero, and since 2022 the estimate is −0.10 pp with a tight interval around zero (Table 3, Figure 3). The same pattern holds overnight. The size of the effect is small in absolute terms. More than 90% of each market's

reaction to the other happens in the same minute, so a lead of 1.9 pp concerns only the last few percent of the adjustment.

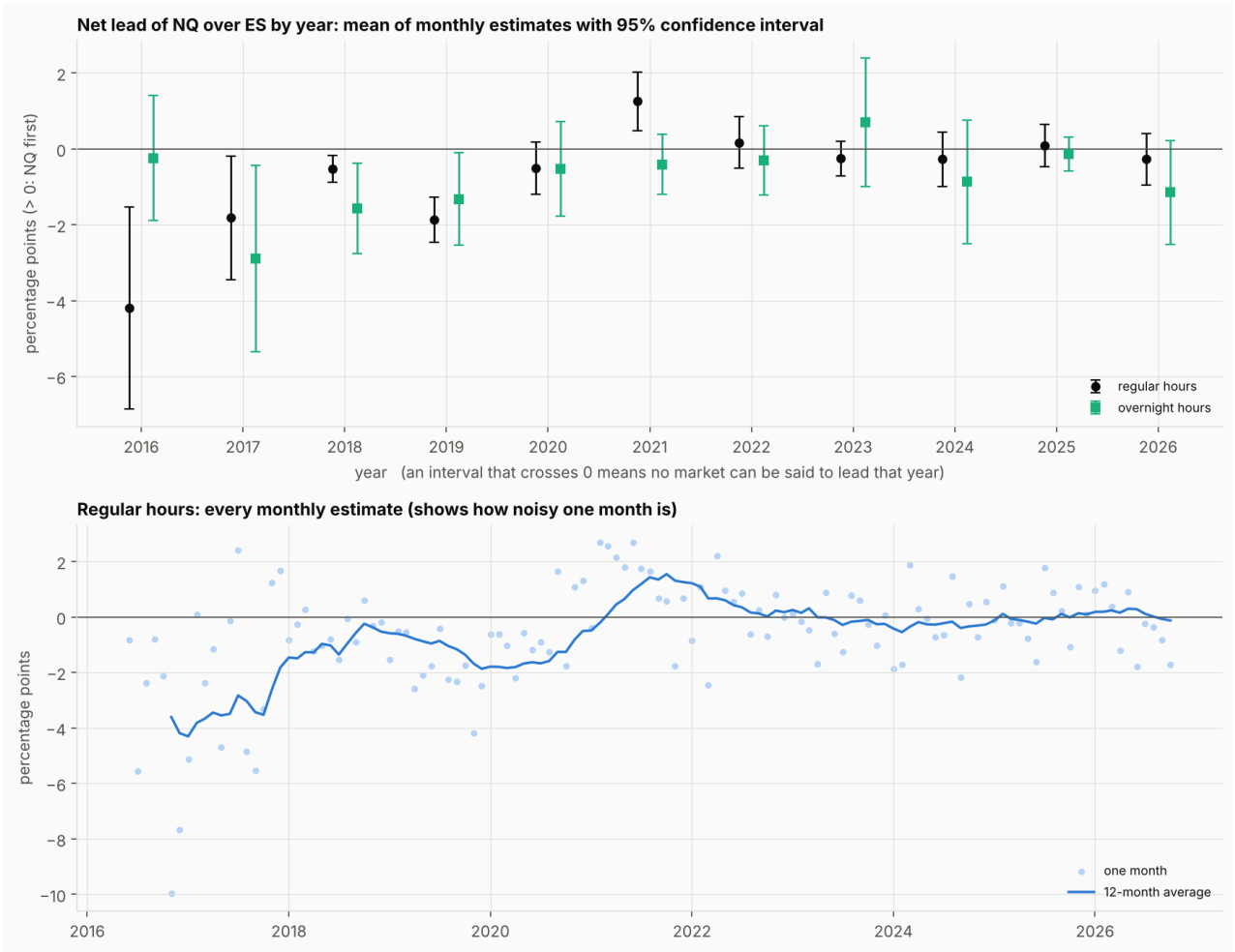



**Figure 3.** Net lead of NQ over ES. Top: annual means of monthly estimates with 95% confidence intervals for regular (black) and overnight (green) hours; an interval that crosses zero means no market can be said to lead that year. Bottom: every monthly estimate for regular hours (dots) and the 12-month average (line). The ES lead of 2016–2019 shrinks towards zero from 2020 onward.

**Liquidity explains the change.** A natural hypothesis is that information shows up first where more trading takes place. NQ's share of NQ + ES dollar volume rose from about 10% in 2016 to about 35% in 2026 (Figure 4, right). Regressing the monthly regular-hours net lead on that share gives a slope of +12.9 (Newey–West t = 4.8, $R^2$ = 0.27, 125 months): each 10-point gain in NQ's volume share adds about 1.3 pp to its net lead. The relation survives a linear time-trend control (slope 18.9, t = 3.9), so it is not simply two series that both trend over time. Overnight, where both markets are thinner, the relation is weaker (slope 4.8, t = 1.9).

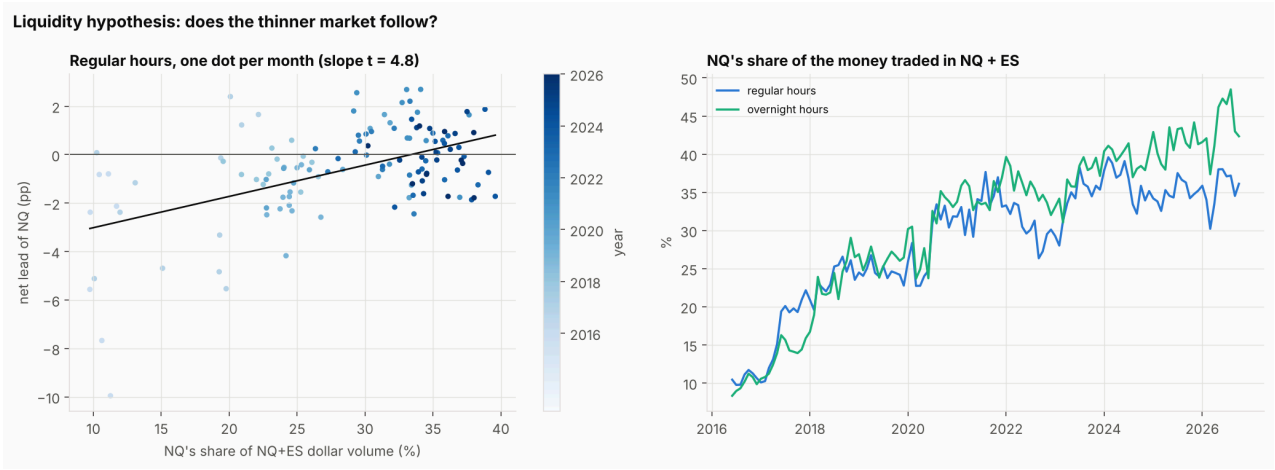



**Figure 4.** Left: monthly net lead of NQ against NQ's share of NQ + ES dollar volume, regular hours, coloured by year, with the OLS line. Right: NQ's share of the dollar volume traded in the two contracts.

## 6. Results II: futures versus ETFs

**Table 4.** Net lead of the future over its ETF (pp), regular hours, mean of monthly estimates with 95% confidence interval.

| Period | NQ vs QQQ | ES vs SPY |
|---|---|---|
| 2016–2019 | **+0.62** [0.44, 0.79] | **+1.82** [1.35, 2.28] |
| 2020–2021 | +0.34 [0.12, 0.56] | +0.46 [0.25, 0.68] |
| 2022–2026 | +0.10 [−0.02, 0.23] | +0.21 [0.01, 0.40] |
| All years | +0.33 [0.22, 0.43] | +0.81 [0.58, 1.04] |

**Futures lead their ETFs, but at one-minute resolution the lead has almost vanished.** The futures move first in every period, consistent with Hasbrouck (2003). The lead was largest for ES over SPY in 2016–2019 (+1.82 pp, t = 7.7) and has shrunk to a fraction of a percentage point since 2022. For NQ over QQQ, the 2022–2026 interval includes zero (Table 4, Figure 5).

**Textbook measures cannot say who leads at this frequency.** On the real data the bounds of Hasbrouck's information share are about [0.01, 0.96] in every year (Figure 5, bottom), so the futures' share is "somewhere between almost nothing and almost everything". The component share falls outside [0, 1] in most months. This is a known weakness: when two prices react within the same bar, the Cholesky ordering rather than the data largely decides the information share, and Putniņš (2013) shows that the component share responds to differences in noise rather than to who leads. The validated net lead does not have this problem.

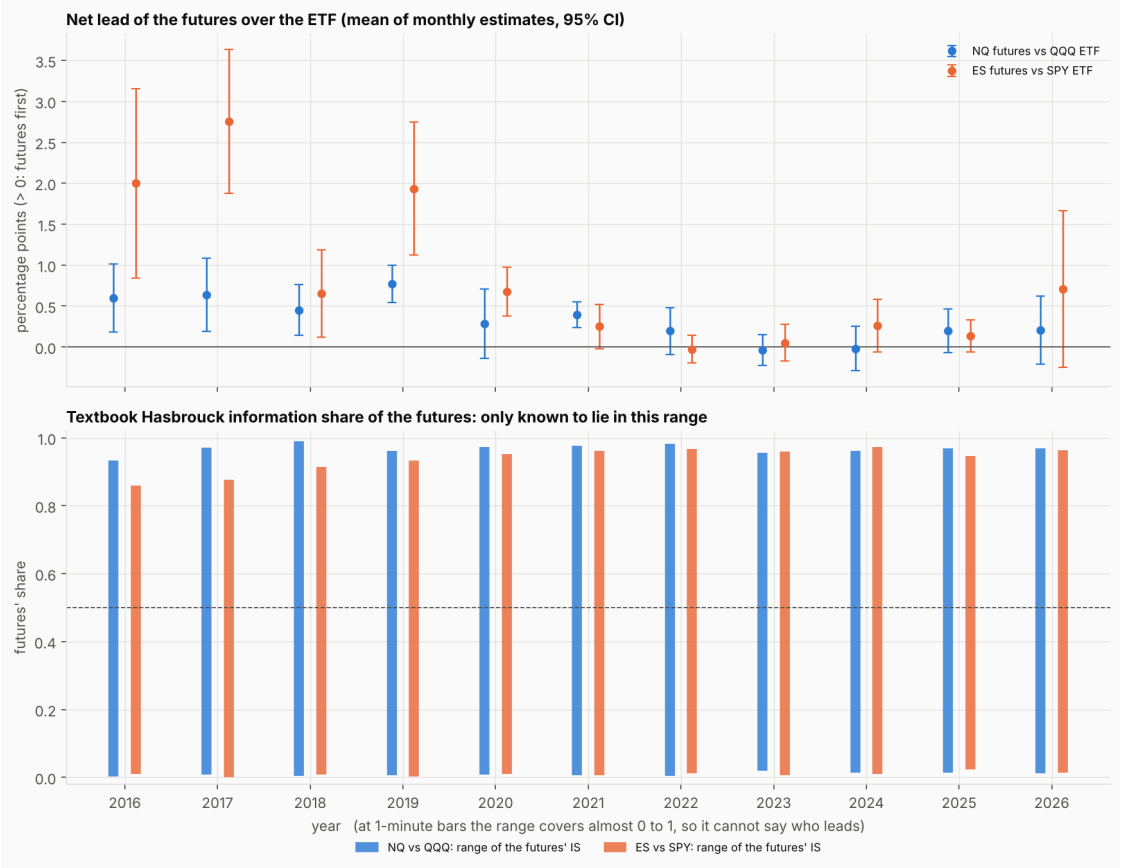



**Figure 5.** Top: annual net lead of the future over the ETF with 95% confidence intervals. Bottom: range of the futures' Hasbrouck information share (lower to upper bound) by year. At one-minute bars the range covers almost 0 to 1, so the measure cannot identify the leader.

## 7. Results III: macro announcements

**Correcting the clock first.** The vendor calendar stamps about 90% of its back-filled releases exactly one hour late (payrolls at 09:30 instead of 08:30 New York, the FOMC at 15:00 instead of 14:00), while entries recorded live in 2026 are correct. Prices settle the question. The biggest one-minute NQ move within ±90 minutes falls on the official release minute for 46.7% of CPI releases and 47.6% of payroll releases, against 4.9% and 8.1% on the vendor's minute. Release dates are therefore taken from the calendar and times from the official schedule (CPI and payrolls 08:30, ISM Manufacturing 10:00, FOMC 14:00). The two unscheduled FOMC cuts of March 2020 are excluded.

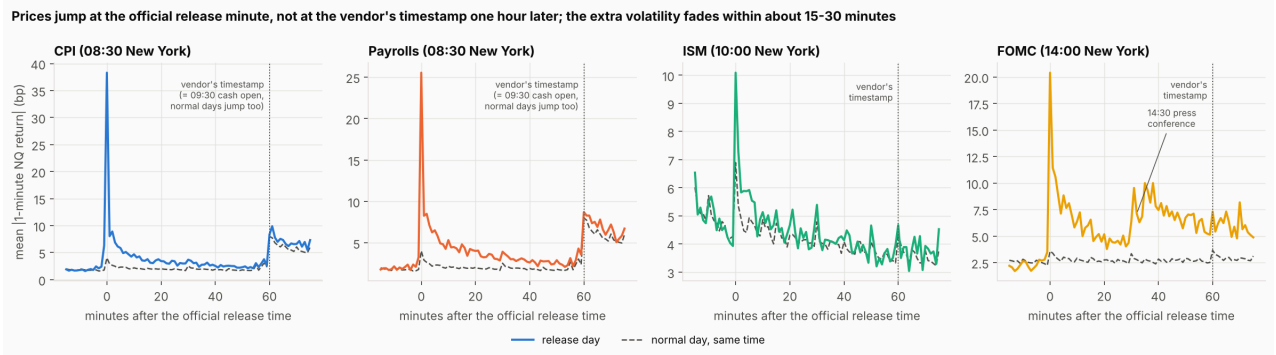



**Figure 6.** Mean absolute one-minute NQ return around each type of release (solid) and at the same clock time on the ten nearest normal days (dashed). Prices jump at the official minute (0), not at the vendor's timestamp an hour later (dotted line, which coincides with the 09:30 stock-market open for the 08:30 releases). FOMC days show a second burst at the 14:30 press conference.

**News moves prices strongly and briefly.** The minute of a CPI release is 9.5 times as volatile as the same minute on a normal day; payrolls 6.3 times and FOMC decisions 5.7 times (ISM, released at 10:00 when markets

are already busy, 1.5 times). The extra volatility fades within 15–30 minutes (Figure 6). Prices follow the surprise, measured as actual minus consensus in standard deviations: each standard deviation of CPI surprise moves ES by −32 bp over five minutes (t = −4.1), while stronger payroll and ISM numbers lift it (+4.4 bp, t = 5.7, and +5.8 bp, t = 3.6). This is consistent with Andersen et al. (2003): prices jump to the new information within minutes.

**Nobody leads on news.** After CPI, payroll and FOMC releases, NQ and ES move more tightly together than at the same time on normal days (same-minute correlation 0.95–0.97 against 0.91–0.93), and every 95% confidence interval of the net lead in the 20 minutes after releases includes zero (bootstrap over releases). Around ISM and FOMC releases, which fall inside the ETFs' regular session, the futures-over-ETF lead also disappears (+0.09 pp [−0.55, 0.62] for NQ vs QQQ). Public, scheduled news reaches all four markets at once. The leads of Sections 5 and 6 come from the rest of the day's information flow: order flow and news that only some traders see first.

## 8. Results IV: one second

**The staleness trap.** A placebo shows how large the fake lead of Section 3.6 is. A copy of NQ that simply trades in fewer seconds, 83% of them like ES, but has no real delay appears to "lag" NQ: its lag-1 asymmetry is +0.062 on all seconds, and the VAR net lead is +7.4 pp, larger than any real lead found at one minute. Restricting the sample to seconds in which both series traded removes the fake lead (−0.0001, 95% CI [−0.0011, 0.0009]). With a copy that trades in only 70% of seconds, a small residual of 0.003 remains. That residual is the benchmark for the real results.

**Table 5.** Lag-1 asymmetry at one second, seconds in which both markets traded, mean of daily estimates (July–October 2026).

| Pair (a vs b) | $\text{corr}(a_t, b_{t+1})$ | $\text{corr}(b_t, a_{t+1})$ | Asymmetry [95% CI] | Days | Days with a first |
|---|---|---|---|---|---|
| NQ vs QQQ | 0.095 | 0.017 | **+0.077** [0.071, 0.084] | 73 | 100% |
| ES vs SPY | 0.087 | 0.044 | **+0.042** [0.038, 0.047] | 73 | 100% |
| NQ vs ES | 0.056 | 0.022 | **+0.035** [0.030, 0.039] | 75 | 91% |

**At one second the leads are clear.** The futures lead their ETFs on every day of the sample, and NQ leads ES on 91% of days (Table 5, Figure 7). The asymmetries are 10–25 times larger than the placebo's residual. They were invisible at one-minute bars (NQ vs ES: −0.10 pp since 2022) because almost the whole reaction happens within the same second and the rest within one or two seconds. The NQ-over-ES lead fits the liquidity story of Section 5: NQ now trades in 89% of regular-hours seconds against ES's 83%.

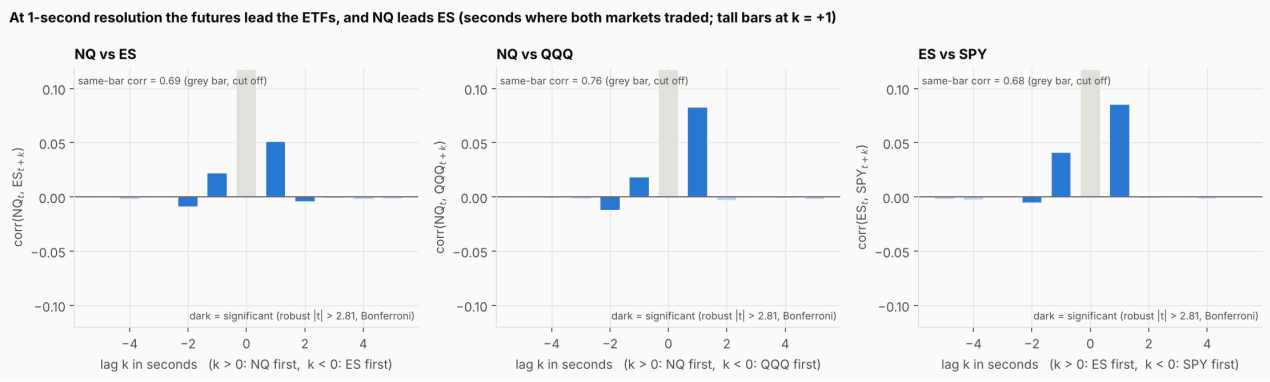



**Figure 7.** Cross-correlation at one-second lags on seconds where both markets traded. Dark bars are significant (robust |t| above the Bonferroni threshold); the same-second bar (grey) is cut off. The tall bars at $k = +1$ show the futures moving before the ETFs and NQ before ES.

## 9. Economic significance

**Table 6.** One-minute speed: hedged catch-up over the next minute after the leader's top-1% moves, mean of monthly estimates (bp).

| Trade | Period | Catch-up [95% CI] | Cheapest cost | Catch-up as % of cost |
|---|---|---|---|---|
| ES moves → trade NQ | 2016–2019 | **0.40** [0.30, 0.49] | 0.73 | 55% |
| | 2022–2026 | 0.05 [−0.07, 0.18] | 0.26 | 21% |
| NQ moves → trade QQQ | 2016–2019 | 0.06 [0.03, 0.09] | 1.13 | 5% |
| | 2022–2026 | 0.09 [0.01, 0.17] | 0.41 | 22% |
| ES moves → trade SPY | 2016–2019 | 0.17 [0.13, 0.21] | 0.66 | 25% |
| | 2022–2026 | 0.05 [0.01, 0.08] | 0.34 | 14% |

1 bp = 0.01%. Cheapest cost: one round trip in the follower at a one-tick spread plus exchange or broker fees, with no market impact.

**Nothing can be traded at one-minute speed.** The leads of Sections 5 and 6 are statistically real (the ES-to-SPY catch-up in 2016–2019 has t = 8.9) but economically too small. Almost all of the follower's reaction happens in the same minute as the leader's move: after an average 12.5 bp ES move in 2016–2019, NQ had already moved 14.4 bp in that minute. What is left to catch is a fraction of a basis point, never more than 55% of the cheapest cost of the trade (Table 6, Figure 8). The unhedged version of the trade, buying the follower alone, is not significantly profitable in any period either.

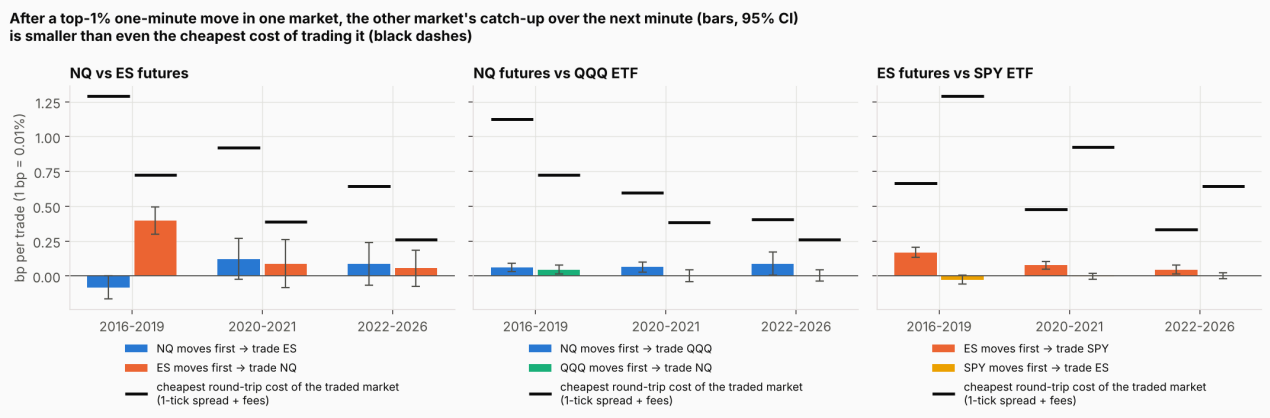



**Figure 8.** Hedged catch-up over the next minute after a top-1% one-minute move (bars, 95% CI) against the cheapest round-trip cost of trading the follower (black lines), by period and direction.

**Table 7.** One-second speed: follower's gain after the leader's top-0.1% seconds (bp), mean of daily estimates.

| Trade (follower only) | Best gain before costs [95% CI] | Best hold | Cheapest cost | Gain if 1 s late |
|---|---|---|---|---|
| NQ moves → trade QQQ | **0.27** [0.10, 0.43] | 2 s | 0.24 | 0.07 |
| ES moves → trade SPY | **0.18** [0.06, 0.29] | 2 s | 0.22 | 0.06 |
| ES moves → trade NQ | 0.10 [0.01, 0.18] | 1 s | 0.15 | −0.03 |

Gains assume trading instantly at the last traded price, with a one-tick spread and no market impact, so real gains would be smaller.

**At one second, the lead is worth about its cost, for about two seconds.** After the futures' biggest seconds, the ETF's catch-up peaks after two seconds at 0.27 bp for QQQ and 0.18 bp for SPY, about the size of the cheapest cost of the trade (Table 7, Figure 9). A trader who reacts one second late keeps only 0.06–0.07 bp, well below cost. A one-minute trader gets nothing, a trader reacting within a second roughly breaks even, and whatever profit exists belongs to firms that react in milliseconds. This is the "arms race" that Budish, Cramton and Shim (2015) document for ES and SPY. It is also exactly what Grossman and Stiglitz (1980) predict: the lead survives only up to the cost of exploiting it.

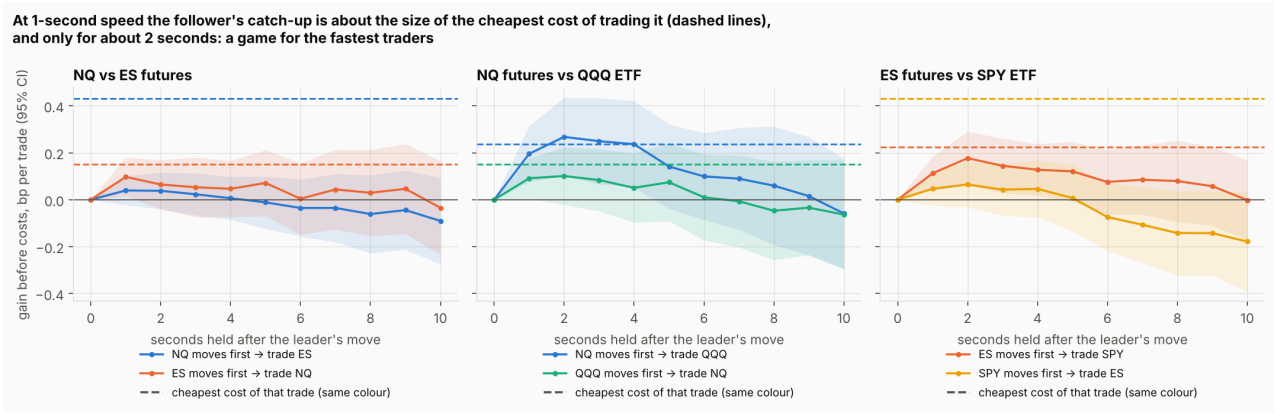



**Figure 9.** Follower's gain before costs (95% CI) by seconds held after the leader's top-0.1% seconds, against the cheapest cost of the same trade (dashed, same colour).

## 10. Robustness and limitations

- **Robustness.** With top-10% or top-0.1% moves instead of top-1%, the one-minute catch-up never exceeds the cheapest cost (at most 71% of it). Aggregating to bars of 2 to 30 minutes does not reveal a larger NQ–ES lead (full-sample estimates between −1.7 and +1.0 pp). Monthly estimation means that a few extreme days cannot drive a period's result.
- **Sub-second timing.** One-second bars cannot separate a lead of a few hundred milliseconds from one market's last trade simply landing later within the second. Millisecond tick data, available from the same vendor, would time the lead exactly.
- **Short one-second sample.** The one-second results cover 73–75 trading days. They are very consistent from day to day, but a longer sample would show whether they change over time.
- **Trade prices, not quotes.** Bars are built from last-trade prices, so gains are measured at prices that may not have been executable, and costs assume a one-tick spread with no market impact. Both choices favour the trader, so the conclusion that the leads cannot be traded is conservative.
- **One vendor.** All data come from London Strategic Edge. Its quality was cross-checked against the ETFs, known closes and news timing, and the two issues found were corrected or documented.

## 11. Conclusion

Information in the US equity-index complex travels fast and almost evenly. At one-minute resolution, the ES lead over NQ that existed in 2016–2019 has disappeared, and its disappearance tracks NQ's rise in liquidity; futures still move before their ETFs, but by a fraction of a percentage point. Scheduled macro news is priced by all four markets within the same minute. At one-second resolution the ordering is clear and stable — futures first, NQ before ES — yet the lead is worth roughly what it costs to trade and lasts about two seconds. The markets are efficient in exactly the sense Grossman and Stiglitz (1980) describe: the remaining predictability pays only the fastest traders for keeping prices aligned.

A second lesson is methodological. Two standard tools, Granger tests and Hasbrouck information shares, gave misleading answers on this data: the first because of tick size, the second because of same-bar reactions. Testing every measure on data with a known answer before using it on real data changed the conclusions of this study, and is cheap to do.

Natural extensions are millisecond tick data, to time the lead precisely, and other assets that trade in several venues at once, such as Treasury futures and their ETFs, or interest-rate expectations priced in both futures and prediction markets.

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## Appendix: reproducibility

The full pipeline is public at github.com/unaifmm/information-diffusion-nasdaq-sp500 (release v1.0). Ten numbered Python scripts run in order, from the data download (`01_download_data.py`) to the economic-significance analysis (`10_economic_significance.py`), and `python run_all.py --download` reproduces every figure and table in this paper. Thirty-one unit tests check the building blocks against cases with a known answer and run automatically on GitHub at every change. Raw data are not included under the vendor's terms. The Python code was written with the help of AI coding assistants; the research questions, validation design and interpretation are the author's, and every result was checked against known answers before it was used. This paper is for research and educational purposes only and is not investment advice.